# JURONG JUNIOR COLLEGE
## JC 2 PRELIMINARY EXAMINATION
**Higher 1**

| | | | |
|---|---|---|---|
| CANDIDATE NAME | | | |
| CLASS | | EXAM INDEX NUMBER | |

# CHEMISTRY **8872/02**

Paper 2 Structured Questions

**30 August 2013**

**2 hours**

Candidates answer Section A on the Question Paper.

Additional Materials: Answer Paper
Data Booklet

**READ THESE INSTRUCTIONS FIRST**

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

**Section A**
Answer **all** questions.

**Section B**
Answer **two** questions on separate answer paper.

A *Data Booklet* is provided. Do not write anything on the *Data Booklet*.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

**For Examiner's Use**

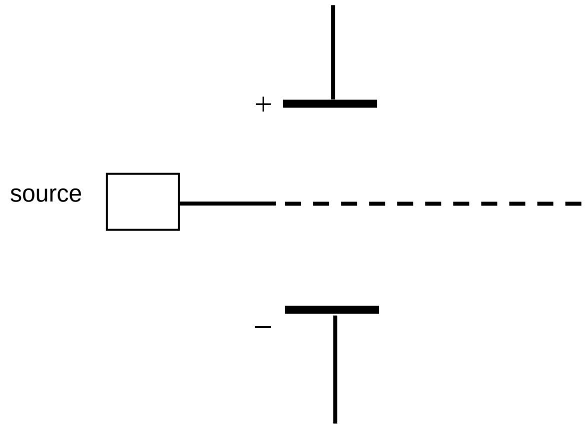
| **Section A** | | **Section B** | |
|---|---|---|---|
| **1** | | **7** | |
| **2** | | **8** | |
| **3** | | **9** | |
| **4** | | | |
| **5** | | | |
| **6** | | | |
| **Total** | | | |

This document consists of **18** printed pages.

## Section A

Answer **all** questions in this section in the spaces provided.

**1.** **(a)** **(i)** When separate beams of $^{14}O^{2-}$ and $^{24}Mg^{2+}$ are passed through an electric field in the set-up shown below, they behave differently.

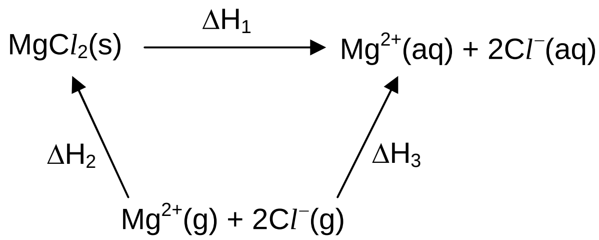



Sketch on the diagram above to show the path of the beam of $^{24}Mg^{2+}$ as it enters the electric field. **Label the beam.**

**(ii)** Given that the angle of deflection for $^{24}Mg^{2+}$ is 2.5$^{o}$, calculate the angle of deflection of the $^{14}O^{2-}$ beam.

[2]

**1.** **(b)**




**(i)** What enthalpy change does $\Delta H_2$ represent?

…………………………………………………………………………………

**(ii)**

| **Reaction** | **Enthalpy change, $\Delta H$** |
|---|---|
| $Mg^{2+}(g) \rightarrow Mg^{2+}(aq)$ | $-1920$ kJ mol$^{-1}$ |
| $Cl^-(g) \rightarrow Cl^-(aq)$ | $-381$ kJ mol$^{-1}$ |
| $Mg^{2+}(g) + 2Cl^-(g) \rightarrow MgCl_2(s)$ | $-2326$ kJ mol$^{-1}$ |

Using the above values, calculate $\Delta H_3$ and hence $\Delta H_1$.

**(iii)** How would you expect the magnitude of the lattice energy of magnesium oxide to compare with magnesium chloride. Explain your answer.

[5]

[Total: 7]

**2.** In aqueous solution, methanoic acid ($HCO_2H$) reacts with bromine ($Br_2$).

$$HCO_2H(aq) + Br_2(aq) \rightarrow 2H^+(aq) + 2Br^-(aq) + CO_2(g)$$

Student **X** carried out an investigation on the rate of this reaction. He uses a large excess of methanoic acid (0.500 mol dm$^{-3}$). During the reaction, bromine is used up and loses its brown colour. The colour intensity of the bromine colour can be measured using a colorimeter to give the concentration of $Br_2$.

| Time/s | 0 | 50 | 100 | 150 | 230 | 300 | 400 |
|---|---|---|---|---|---|---|---|
| [$Br_2$(aq)]/ mol dm$^{-3}$ | 0.0100 | 0.0080 | 0.0067 | 0.0055 | 0.0040 | 0.0030 | 0.0022 |

**(a)** Explain the purpose of using large excess of methanoic acid.

[1]

**(b)** Using the graph paper on the next page, plot a graph and determine the order of reaction with respect to bromine.

[3]

**(c)** Given that the reaction is first order with respect to methanoic acid, write the rate equation for the reaction.

[1]

**(d)** Determine the initial rate of the experiment carried out by Student **X**. Hence calculate the rate constant, stating its units. You should show your clear working on your graph.

[2]

**2.** **(e)** Student **Y** also carried out an experiment to study the reaction between methanoic acid ($HCO_2H$) and bromine ($Br_2$).

$$HCO_2H(aq) + Br_2(aq) \rightarrow 2H^+(aq) + 2Br^-(aq) + CO_2(g)$$

He reacted 50 $cm^3$ of 0.500 mol $dm^{-3}$ $HCO_2H$ with 50 $cm^3$ of 0.500 mol $dm^{-3}$ $Br_2$ at 25°C and measured the volume of gas collected every 10 minutes. Then he plotted Curve **R** in **Figure 2.1**.

Sketch on **Figure 2.1** to show how the volume of $CO_2$ collected changes with time if he repeated the experiment using 50 $cm^3$ of 0.500 mol $dm^{-3}$ $HCO_2H$ with 50 $cm^3$ of 0.500 mol $dm^{-3}$ $Br_2$ at 100°C.

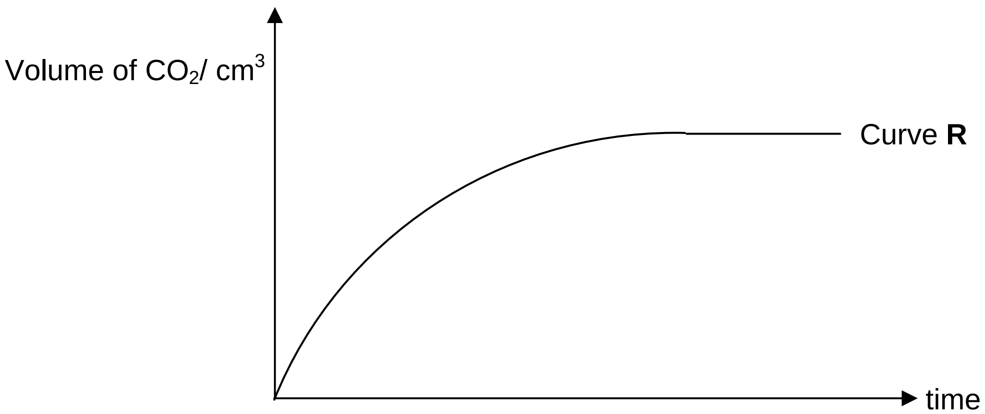
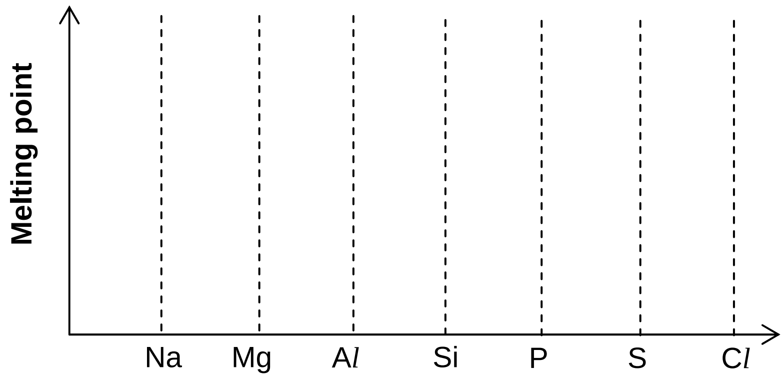


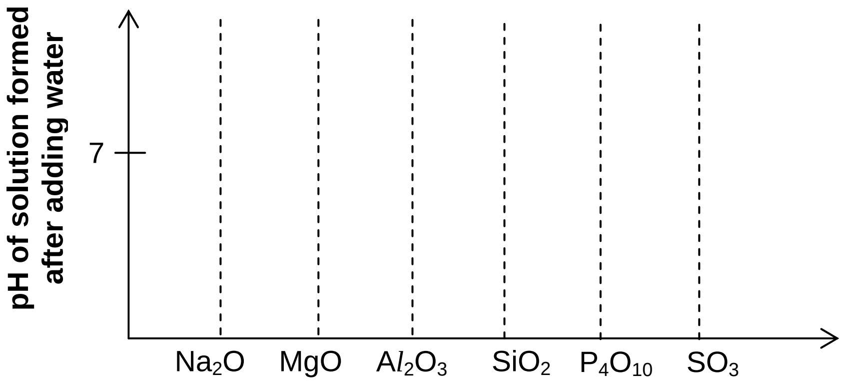

**Figure 2.1**

[1]

[Total: 8]

**3.** **(a)** On the following diagrams, sketch graphs to show the variation in the named property of some Period 3 elements or compounds.




State the pH values formed by each oxide on your sketch. [2]

**(b)** Explain why sodium and magnesium have different melting point.

[2]

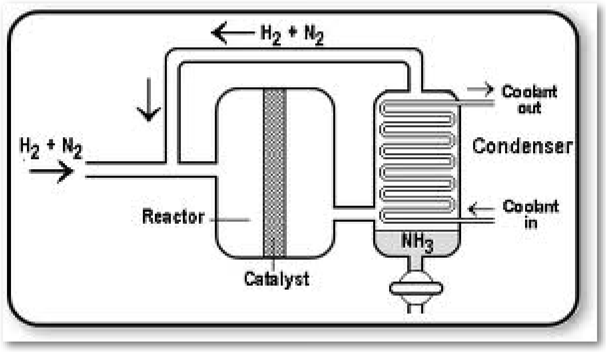
**(c)** Write equations to explain the difference in the pH of the solution formed after water is added to $P_4O_{10}$ and MgO.

[2]

[Total: 6]

**4.** Haber process is an industrial process to manufacture ammonia from nitrogen.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$




**(a)** Give two features of a reaction at equilibrium.

Feature 1:

Feature 2:

[2]

**(b)** **(i)** What are the typical values of pressure and temperature used industrially?

Temperature ………………………………………….

Pressure …………………………………………….

**(ii)** Explain why that particular temperature in **4b(i)** is chosen.

[3]

**4.** **(c)** **(i)** Sketch a graph on the axes given below to show how the % yield of $NH_3$ changes with pressure.


**(ii)** Explain the shape of your graph in **4(c)(i)**.

[2]

[Total: 6]

**5.** The **Dissolved Oxygen Concentration** (DOC) in rivers and lakes is important for aquatic life. If the DOC falls below 5 ppm, most species of fish cannot survive.

Environmental chemists can determine the DOC in water using the procedure below.

- **Step 1**: A sample of river water is shaken with aqueous $Mn^{2+}$ and aqueous alkali. The dissolved oxygen oxidises the $Mn^{2+}$ to $Mn^{3+}$, forming a pale brown precipitate of $Mn(OH)_3$.

$$O_2(aq) + 4Mn^{2+}(aq) + 8OH^-(aq) + 2H_2O(l) \rightarrow 4Mn(OH)_3(s)$$

- **Step 2**: $Mn(OH)_3$ precipitate is then reacted with an excess of aqueous potassium iodide, which is oxidised to iodine.

$$Mn(OH)_3(s) + e^- \rightleftharpoons Mn(OH)_2(s) + OH^-(aq)$$
$$I_2(aq) + 2e^- \rightleftharpoons 2I^-(aq)$$

- **Step 3**: The iodine formed is then determined by titration with aqueous sodium thiosulfate, $Na_2S_2O_3(aq)$.

$$2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

25.0 $cm^3$ of a sample of river water was analysed using the above procedure. The titration requires 24.60 $cm^3$ of 0.00100 mol $dm^{-3}$ $Na_2S_2O_3(aq)$.

**(a)** Calculate the amount of iodine reacted during the titration in **Step 3**.

[1]

**(b)** Using the two half equations given above in **Step 2**, construct a balanced equation for the reaction that occurred in **Step 2** of the procedure.

[1]

**(c)** Using the equation you constructed in **(b)**, calculate the amount of $Mn(OH)_3$ reacted in **Step 2**.

[1]

**(d)** Hence calculate the amount of $O_2(aq)$ in 25.0$cm^3$ of the sample of river water.

[1]

**5.** **(e)** Calculate the **Dissolved Oxygen Concentration** (DOC), in g $dm^{-3}$, in the river water.

[1]

**(f)** Assume the density of the river water is 1 g $cm^{-3}$, calculate the **Dissolved Oxygen Concentration** (DOC), in ppm, in the river water.

[1 ppm of dissolved oxygen = 1 g of dissolved oxygen in $10^6$ g of river water]

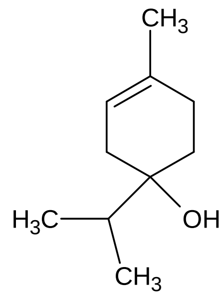
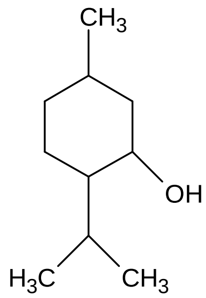
[1]

**(f)** Hence. comment on whether there is enough oxygen for fish to survive in that river.

[1]

[Total: 7]

**6.** Shown below are some active ingredients in essential oils.

| | |
|---|---|
|  |  |
| Terpinen-4-ol<br>(Found in lavender essential oil) | Menthol<br>(Found in peppermint essential oil) |

**(a)** Identify all the functional groups in terpinen-4-ol.

……………………………………………………………………………………………….. [1]

**(b)** **(i)** When subjected to dehydration, terpinen-4-ol forms $x$ isomeric organic products while menthol forms $y$ isomeric organic products.

What are the numbers $x$ and $y$?

$x$ = ……………………………  $y$ = …………………………………..

**(ii)** State the reagents and conditions required for the dehydration.

…………………………………………………………………………………….

**(iii)** Draw the structural formula of one organic product that is obtained when menthol undergoes dehydration.

[3]

**(c)** Both terpinen-4-ol and menthol decolourise purple $KMnO_4$. Draw the structural formula of the products formed.

Oxidation product of terpinen-4-ol:

Oxidation product of menthol:

[2]

**[Total:6]**

## Section B

Answer **two** questions from this section on separate answer paper.

**7.** **(a)** Burning of fossil fuels emitted large amounts of sulfur oxides ($SO_x$) and nitrogen oxides ($NO_y$). When exposed to the atmosphere, these oxides react with water to form sulfuric acid and nitric acid which are the components of acid rain.

**(i)** In the atmosphere, nitric oxide (NO) is oxidised by ozone ($O_3$) to form nitrogen dioxide ($NO_2$), which in turn reacts with •OH radical to form nitric acid ($HNO_3$).

Draw dot-and-cross diagrams for $NO_2$ molecule and •OH radical.

Hence explain why they react readily to form nitric acid ($HNO_3$).

**(ii)** Besides forming nitric acid, NO contributes to acid rain by being a homogeneous *catalyst* in the oxidation of $SO_2$ to $SO_3$.

The catalysed reaction occur in two steps:

$$NO(g) + \tfrac{1}{2}O_2(g) \rightarrow NO_2(g) \qquad \Delta H = -57 \text{ kJ mol}^{-1}$$

$$NO_2(g) + SO_2(g) \rightarrow SO_3(g) + NO(g) \qquad \Delta H = -88 \text{ kJ mol}^{-1}$$

The activation energy of the first step, $E_{a1}$, is lower than that of second step, $E_{a2}$.

Use the information above to construct a fully-labelled reaction pathway diagram for this two-step reaction.

**(iii)** Explain the term "*catalyst*". [7]

**(b)** A sample of carbon disulphide, $CS_2$, reacts with excess nitrogen monoxide, NO, to form a yellow residue, sulfur, and two gases $CO_2$ and $N_2$. The gaseous products occupied a total volume of 130 $cm^3$. However, when passed through aqueous sodium hydroxide, the volume of the gas decreased.

All gases were measured at room temperature.

**(i)** Write a balanced equation for the reaction between $CS_2$ and NO.

**(ii)** Calculate the mass of carbon disulphide, $CS_2$, in the sample.

**(iii)** Write an equation to show why the volume of gaseous product decreases when passed through aqueous sodium hydroxide.

**(iv)** Describe, in terms of orbital overlap, the bonding between sulfur and carbon atoms in a $CS_2$ molecule. A clearly labelled diagram may clarify your answer. [6]

**7. (c)** Both chloroethane and iodoethane react with $NH_3$ in ethanol under pressure to give ethylamine.

$$CH_3CH_2Cl \xrightarrow[\text{Heat in sealed tube}]{NH_3 \text{ in ethanol}} \underset{\textbf{\textit{ethylamine}}}{CH_3CH_2NH_2} \xleftarrow[\text{Heat in sealed tube}]{NH_3 \text{ in ethanol}} CH_3CH_2I$$

**(i)** What conditions must be used in the above reaction so that ethylamine can be obtained as a major product?

**(ii)** In the above reaction with $NH_3$, is $CH_3CH_2Cl$ more or less reactive compared to $CH_3CH_2I$?

Explain your answer. [3]

**(d)** Chloroethane can also be converted to propylamine.

$$CH_3CH_2Cl \xrightarrow{\text{Step } \mathbf{I}} \mathbf{X} \xrightarrow{\text{Step } \mathbf{II}} \underset{\textbf{\textit{propylamine}}}{CH_3CH_2CH_2NH_2}$$

**(i)** State the reagents and conditions needed for Step **I** and **II**.

**(ii)** Give the structural formula of **X**. [3]

**(e)** Chloroethane reacts with methylamine, $CH_3NH_2$, to form compound **Y**, which is an isomer of propylamine.

$$CH_3CH_2Cl \xrightarrow[\text{Heat in sealed tube}]{CH_3NH_2 \text{ in ethanol}} \mathbf{Y}$$

Give the displayed formula of **Y**. [1]

[Total: 20]

**8.** This question is about ethanoic acid and two other naturally occurring carboxylic acids.

**(a)** Some information about two of the processes used to make ethanoic acid is given below.

| **Process 1**: |
| --- |
| This is a one-step process that involves the reaction of methanol with carbon monoxide. $$CH_3OH + CO \rightarrow CH_3COOH$$ The conditions used are 180 °C and 30 atmospheres pressure and a rhodium/iodine catalyst is used. The percentage yield for this process is 99%. |

| **Process 2**: |
| --- |
| This involves the oxidation of naphtha, a fraction obtained from crude oil. Liquid naphtha is oxidised using air at a temperature of 180 °C and 50 atmospheres pressure. No catalyst is needed. A large variety of other products are also formed in this oxidation. |

**(i)** Suggest one advantage of making ethanoic acid using **Process 1** rather than **Process 2**.

**(ii)** Other products formed in **Process 2** are carboxylic acids, aldehydes and ketones.

One such product is compound **L**.

One mole of **L**, $C_3H_4O_3$, forms half a mole of carbon dioxide when reacted with aqueous sodium carbonate. **L** decolourises acidified $KMnO_4$ and forms red-brown precipitate with Fehling's solution.

Draw the structural formulae of **L** and the product formed when **L** reacts with Fehling's solution. [3]

**(b)** In gaseous state just above boiling point, the monomer and dimer forms of ethanoic acid exist together in equilibrium.

$$2CH_3COOH(g) \rightleftharpoons (CH_3COOH)_2$$

Draw a fully-labelled diagram to illustrate the bond formed when ethanoic acid dimerise. [1]

**8.** **(c)** p$K_a$ values of some naturally occurring *Bronsted-Lowry acids* are shown in **Table 8.1** below.

| Common name | IUPAC name | Structural Formula | p$K_a$ value |
|---|---|---|---|
| Acetic acid (from vinegar) | Ethanoic acid | $CH_3CO_2H$ | 4.76 |
| pyruvic acid (formed during metabolism | 2-oxopropanoic acid | $CH_3COCO_2H$ | 2.39 |
| Lactic acid (from milk) | 2-hydroxypropanoic acid | $CH_3CH(OH)CO_2H$ | 3.86 |

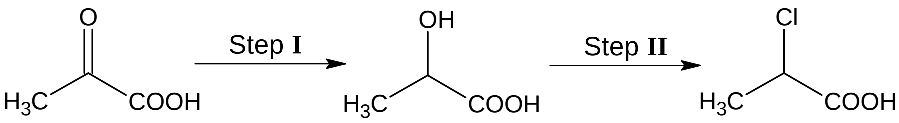
**Table 8.1**

**(i)** What is meant by the term *Bronsted-Lowry acid*?

**(ii)** Arrange in increasing order the relative strength of the three acids in **Table 8.1**.

**(iii)** Explain why pyruvic acid has a lower p$K_a$ than ethanoic acid.

**(iv)** Pyruvic acid can be converted to 2-chloropropanoic acid using the following two-step synthesis.




pyruvic acid 2-chloropropanoic acid

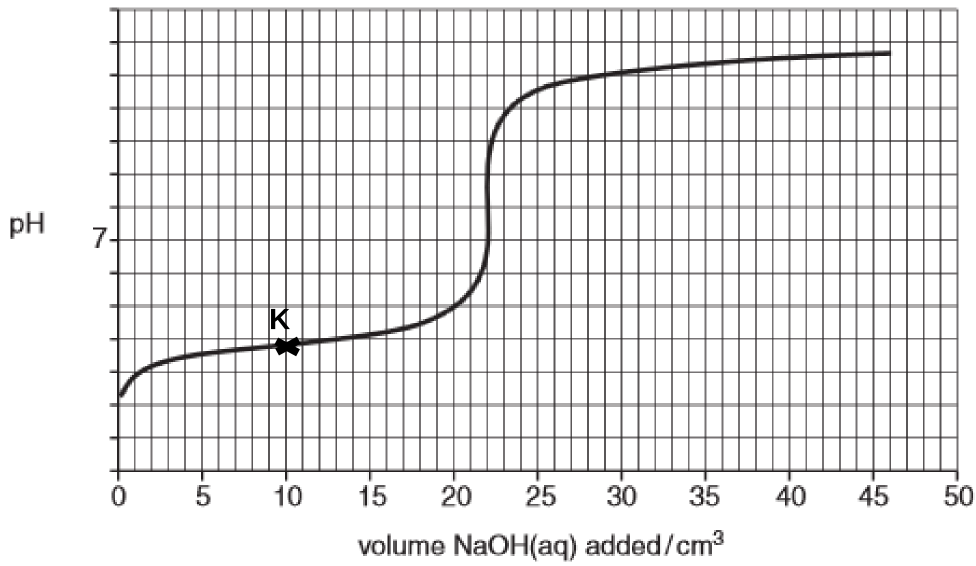
For both Step **I** and **II**, state the type of reaction occurred and the reagents and conditions required. [7]

**(d)** **(i)** Define standard enthalpy change of neutralisation.

**(ii)** Explain why the standard enthalpy change of neutralisation of ethanoic acid is less exothermic than that of HC*l*. [2]

**8.** **(e)** A student pipetted 25.0 $cm^3$ 0.250 mol $dm^{-3}$ of lactic acid into a conical flask. He added NaOH(aq) solution from a burette and monitor the pH of the reaction mixture in the conical flask using a pH meter.

The pH curve obtained by the student is shown below.




**(i)** Using the $pK_a$ value in **Table 8.1**, determine the value of $K_a$ of lactic acid.

Hence calculate the initial pH of the lactic acid in the conical flask before NaOH(aq) was added.

**(ii)** Calculate the concentration of the NaOH(aq) solution.

**(iii)** At point **K**, the solution in the conical flask contains a mixture of lactic acid and its sodium salt. This mixture can be used as buffer.

What do you understand by the term *buffer*? Using a balanced equation, show how the above buffer works when some dilute HC*l* is added to it.

**(iv)** Choose the most suitable indicator for the titration from the table below. State the colour change of the solution at endpoint.

| Indicator | pH at which colour changes | Acid colour | Base colour |
|---|---|---|---|
| Congo red | 3 – 5 | blue | red |
| Brilliant yellow | 6 – 8 | yellow | orange |
| Thymol blue | 8 – 10 | yellow | blue |

[7]

[Total: 20]

**9.** **(a)** **P**, **Q** and **R** are three structural isomers with the molecular formula of $C_8H_8O_2$.

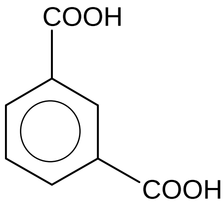
**P** and **Q** react with aqueous NaOH at room temperature.

**R** does not react with aqueous NaOH at room temperature but when heated with aqueous NaOH, **R** forms compound **S**, $C_7H_5O_2Na$ and compound **T** which has a $M_r$ of 32.

**T** forms white fumes with $PCl_5$.

When heated with acidified $KMnO_4$, **P** and **R** oxidised to form the same product, benzoic acid.

However, when heated with acidified $KMnO_4$, **Q** formed 1,3-benzene dicarboxylic acid.




1,3-benzene dicarboxylic acid

**(i)** Suggest structures of **P**, **Q**, **R**, **S**, and **T**. Show how you deduce these structures and suggest the types of reactions that occur.

**(ii)** Write balanced equation for the following reactions:

- **T** reacts with $PCl_5$.
- **Q** reacts with hot acidified $KMnO_4$. [11]

**(b)** Phosphorus reacts with chlorine to form liquid $PCl_3$ and solid $PCl_5$. Unlike phosphorus, nitrogen only forms $NCl_3$.

**(i)** Explain why phosphorus can form $PCl_5$ but nitrogen does not form $NCl_5$.

**(ii)** By considering the number of bond pairs and lone pairs around P, predict and explain bond angle in $PCl_3$.

**(iii)** Explain why $PCl_5$ is a solid but $PCl_3$ is a liquid. [6]

**(c)** Student **G** added a few drops of water to $PCl_5$ solid. To the same amount of $PCl_5$ solid, student **H** added excess cold water which contains a few drops of universal indicator.

**(i)** Student **G** and **H** make different observations.

Describe what does student **G** and **H** see.

**(ii)** Write a balanced equation, with state symbols, for the reaction that student **G** carried out. [3]

[Total: 20]